Mushfiqur Rahman Pulok

Dhaka, Bangladesh

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Summary

I am Mushfiqur Rahman Pulok, a graduate student from Bangladesh University of Business and Technology with hands-on experience in Artificial Intelligence, Machine Learning, and Deep Learning. I enjoy building models, experimenting with architectures, and solving real-world problems through data-driven approaches

EDUCATION

Bangladesh University of Business and Technology, Mirpur-2, Dhaka-1216 **Jan 2022 – April 2026**
B.Sc. in Computer Science and Engineering — ***CGPA: 3.99/4.00***

B.A.F Shaheen College, Dhaka **Passing Year: 2020**
Higher Secondary Certificate (HSC) — ***GPA: 5.00/5.00***

Rupnagar Govt. Secondary School **Passing Year: 2018**
Secondary School Certificate (SSC) — ***GPA: 5.00/5.00***

Projects

Gesture-Controlled Virtual Smart Whiteboard GitHub Link
Built a webcam-based whiteboard controlled entirely by hand gestures using MediaPipe and OpenCV; supports freehand drawing, smart shape correction, select/move/scale, and paint-bucket fill with no mouse or keyboard.

Facial Emotion Recognition (Real-Time, Browser-Based) GitHub Link
Trained a MobileNetV2 + SE-block model on the RAF-DB dataset to classify 7 emotions and exported it to ONNX for fully client-side, real-time webcam inference in the browser via onnxruntime-web.

FinTech Fraud Detection Using Graph Neural Networks (Accepted at ICICV 2026)
Developed a session-based fraud detection approach using GNNs; building transaction graphs and applying unsupervised anomaly detection to identify fraudulent behavior. Compared Graph Neural Networks with traditional Machine Learning models in fintech fraud detection.

Generalized Fundus Disease Classification across Multi-Dataset Retinal Images
Built a generalized retinal disease classifier by combining multiple fundus datasets and validating on external data, using PyTorch and OpenCV to handle domain shift and dataset harmonization.

Art Style Classification using CNN–Transformer Ensemble
Built a CNN–Transformer ensemble (EfficientNetV2-S, ConvNeXt-Tiny, DINOv2-S) in PyTorch for art style classification on the WikiArt dataset.

Diabetes Prediction via Federated Learning
Designed a federated learning framework for diabetes prediction on Non-IID client data; experimenting with FL algorithms to improve global model accuracy while preserving data privacy.

Crop Disease Detection Using Deep Learning GitHub Link
Developed a multi-class crop disease classification system using PyTorch; applied preprocessing, augmentation, transfer learning (ResNet50, EfficientNet), and Optuna-based tuning to achieve high accuracy across 15 disease categories.

Interactive Underwater World Simulation GitHub Link
Created a real-time underwater world using C++ and OpenGL; implemented animated fish, plants, bubbles, and lighting effects for an interactive academic graphics project.

TECHNICAL SKILLS

Programming: Python, C++
Artificial Intelligence: Computer Vision, Deep Learning (PyTorch).
Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, ML pipelines.
Tools & Version Control: Git, GitHub, Jupyter Notebook, VS Code, Kaggle.

CERTIFICATIONS

- BUBT TAFE Data Analyst Certificate

Achievements

- **Research Paper Accepted at ICICV 2026** — *FinTech Fraud Detection Using Graph Neural Networks.*
- Brainstorming Week — Research Poster (SE): Participation — BUBT.
- Brainstorming Week — Software Development (CSE 300, SkillXchange): Participation — BUBT.

References

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